Subject: Project Assignment: Data & ML Layer Integration

Assigning From: Shailesh Sakariya, C.T.O
Assigned To: Akshit Sonani, Data Science Intern

1. Project Context & Objective
Prelax Infotech is currently developing a large-scale Enterprise Resource Planning (ERP) platform for the agricultural sector. As a core component of this ecosystem, the intern has been assigned to design, develop, and integrate the "Agri-Chain Module."
This module serves as the "Intelligence Layer**"** of the application. Its primary objective is to provide data-driven predictive capabilities, enabling the company's end-users (farmers and agri-businesses) to make scientific sowing and selling decisions.

2. Module Scope: Agri-Chain
Agri-Chain is a specialized Machine Learning and Data Science module. This module designed to be consumed via FastAPI endpoints by the company's primary Android and Web applications.

3. Core Functional Responsibilities
The organization has defined four critical technical pillars for this assignment:

- **Data Engineering (ETL):** Building automated pipelines to synchronize the internal database with external sources such as **Agmarknet** and **IMD Weather systems**.
- **Geopolitical Impact Analytics:** Developing a logic-layer to calculate how external shocks (e.g., Global Conflicts, Crude Oil fluctuations) impact local crop pricing.
- **Predictive Modeling (Yield & Price):** Implementation of **XGBoost** and **Time-Series (Prophet/ARIMA)** models to provide accurate market forecasts.
- **Strategic Advisory Engine:** AI-driven recommendation logic analyzing historical ROI to suggest optimal crop rotation for upcoming seasons.

4. Technical Integration Workflow
To meet industry standards, the following workflow is being implemented:

1. **Research Phase:** Statistical analysis of 5-year historical agricultural data.
2. **Development Phase:** Building the end-to-end ML Pipeline in Python.
3. **Deployment Phase:** Wrapping models in **FastAPI** to facilitate seamless JSON data exchange with Android/Web developers.
4. **Monitoring Phase:** Creating an internal **Streamlit Dashboard** to track model performance.

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